

SECURITY

Methods to encrypt & decrypt data

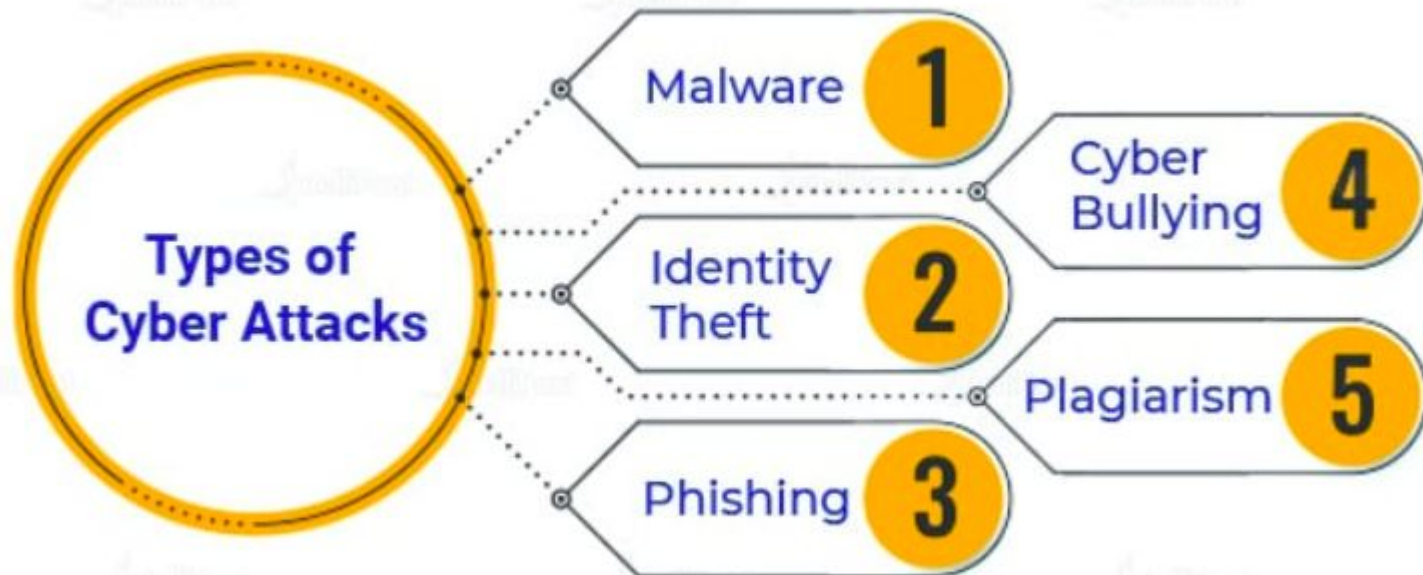
What is Cyber Safety?

Cybersafety addresses the ability to act in a safe and responsible manner on the Internet and other connected environments. These behaviours protect personal information and reputation and include safe practices to minimise danger online.

Why is Cyber Safety Important?

One of the prime reasons for implementing Cyber Safety is the protection of confidential data. Cyber Safety can protect these data to a great extent. These data protection go a long way, especially in cases of government-related data. Breach of such data of national importance can cause serious disturbance to the nation.

Additionally, a breach of personal data would result in personal losses such as reputation damage, etc. There are high chances of an extortion threat. By giving in to the threat, there are chances of financial damages.





Cyber Safety Tips

- Create a strong password for all platforms. Try to create unique passwords. Most importantly, the passwords are not related at all. In this way, hackers will not succeed in their password guessing methodology.
- Always read the terms and conditions of any site. In this way, you will know whether your information will be collected or not.
- Never type your financial information on any site, especially, if the site lacks padlock on the page.
- Always ensure that the permissions required by the particular site are reasonable. For example, if your calculator app requests permission to access your gallery/ any of such nature, it is extremely suspicious. In such cases, uninstall the app immediately.
- Always turn on two-factor authentication on your social media platforms. In that way, you will be alerted whenever there is an attempt to access your account.
- Always backup your data. By doing that you can protect your data in case of breach too.

What is Cryptography?

Cryptography is used to secure and protect data during communication. It is helpful to prevent unauthorized person or group of users from accessing any confidential data.

Cryptography



Cryptography

Cryptography includes following important terms:

- **Plain Text**- It is the original text on sender's side.
- **Cipher Text**-It is the encoded text of the main text.
- **Encryption or Enciphering**-The procedure of transforming clear text to encrypted text is Encryption.
- **Decryption**-The method of changing secret message code back into ordinary readable form of message is Decryption.

Encryption and **Decryption** are the two essential functionalities of cryptography.

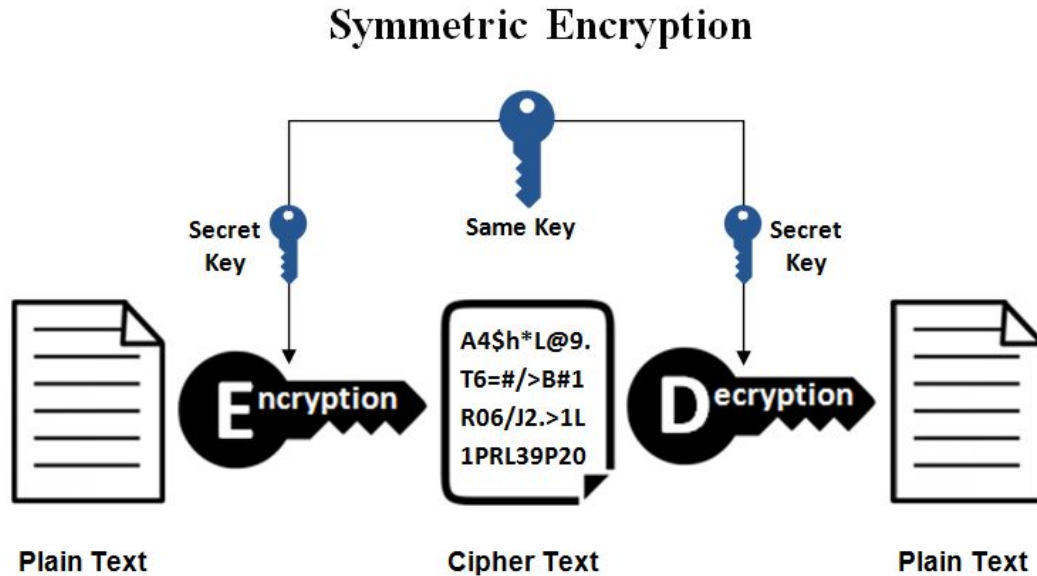
Encryption is a process which transforms the original information into an unrecognizable form. This new form of the message is entirely different from the original message. That's why a hacker is not able to read the data as senders use an encryption algorithm. Encryption is usually done using key algorithms.



Decryption is a process of converting encoded/encrypted data in a form that is readable and understood by a human or a computer. This method is performed by un-encrypting the text manually or by using keys used to encrypt the original data.

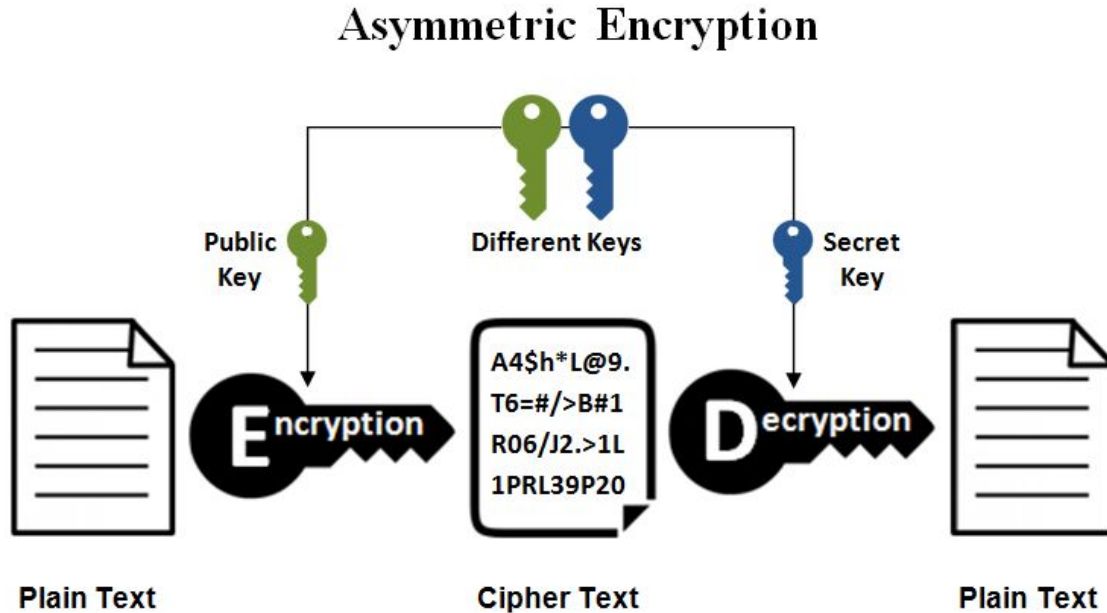


Symmetrical Encryption



This is the simplest kind of encryption that involves only one secret key to cipher and decipher information. Symmetrical encryption is an old and best-known technique. It uses a secret key that can either be a number, a word or a string of random letters. Blowfish, AES, RC4, DES, RC5, and RC6 are examples of symmetric encryption. The most widely used symmetric algorithm is AES-128, AES-192, and AES-256.

Asymmetrical Encryption



Asymmetrical encryption is also known as public key cryptography, which is a relatively new method, compared to symmetric encryption. Asymmetric encryption uses two keys to encrypt a plain text.

Asymmetric encryption is mostly used in day-to-day communication channels, especially over the Internet. Popular asymmetric key encryption algorithm includes ElGamal, [RSA](#), [DSA](#), [Elliptic curve techniques](#), PKCS.

Let's say you were sending a coded message to a friend:

Dear Karen,

I hope you are well. Are we still having dinner tomorrow?

Yours sincerely,

James

Let's say that you coded the message in a simple way, by changing each letter to the one that follows it in the alphabet. This would change the message to:

Efbs Lbsfo,

*J ipqf zpv bsf xfmm. Bsf xf tujmm ibwjoh ejoofs
upnpsspx?*

Zpvst tjodsfmz,

Kbnft

If your enemies intercepted this letter, there is a trick that they could use to try and crack the code. They could **look at the format of your letter and try to guess what the message might be saying**. They know that people normally begin their letters with “Hi”, “Hello”, “Dear” or a number of other conventions.

If they tried to apply “Hi” or “Hello” as the first word, they would see that it wouldn’t fit the number of characters. They could then try “Dear”. It fits, but that doesn’t necessarily mean anything. The attackers would just try it and see where it led them. So they would change the letters “e”, “f”, “b”, and “s” with “d”, “e”, “a”, and “r” respectively. This would give them:

Dear Laseo,

J ipqe zpv are xemm. Are xe tujmm iawjoh djooes upnpsspx?

Zpvrt tjoderemz,

Kanet

It still looks pretty confusing, so the attackers might try looking at some other conventions, **like how we conclude our letters**. People often add “From” or “Kind regards” at the end, but neither of these fit the format. Instead, the attackers might try “Yours sincerely” and replace the other letters to see where it gets them. By changing “z”, “p”, “v”, “t”, “j” “o”, “d” and “m” with “y”, “o”, “u”, “s”, “i”, “n”, “c” and “l” respectively, they would get:

Dear Lasen,

I ioqe you are xell. Are xe tuill iawinh dinnes uonossox?

Yours sincerely,

Kanet

Seeing as the words are in correct grammatical order, the attackers can be pretty confident that they are heading in the right direction. By now, they have probably also realized that the code involved each letter being changed to the one that follows it in the alphabet. **Once they realize this, it makes it easy to translate the rest and read the original message.**

RSA

Where is RSA encryption used?

RSA encryption is often used in **combination with other encryption schemes**, or for **digital signatures** which can prove the authenticity and integrity of a message. It isn't generally used to encrypt entire messages or files, because it is less efficient and more resource-heavy than symmetric-key encryption.

To make things more efficient, **a file will generally be encrypted with a symmetric-key algorithm, and then the symmetric key will be encrypted with RSA encryption.** Under this process, only an entity that has access to the RSA private key will be able to decrypt the symmetric key.

RSA is still seen in a range of web browsers, email, VPNs, chat and other communication channels. RSA is also often used to make secure connections between VPN clients and VPN servers.

RSA

RSA uses much larger numbers. The size of the primes in a real RSA implementation varies, but in 2048-bit RSA, they would come together to make keys that are 617 digits long. To help you visualize it, a key would be a number of this size:

[illegible]

RSA

The reality is that all of the information that our computers process is stored in binary (1s and 0s) and we use encoding standards like **ASCII or Unicode** to represent them in ways that humans can understand (letters).

This means that **keys like “n38cb29fkbjh138g7fqijnf3kaj84f8b9f...” and messages like “buy me a sandwich” already exist as numbers**, which can easily be computed in the RSA algorithm. The numbers that they are represented by are much larger and harder for us to manage, which is why we prefer to deal with alphanumeric characters rather than a jumble of binary.

RSA

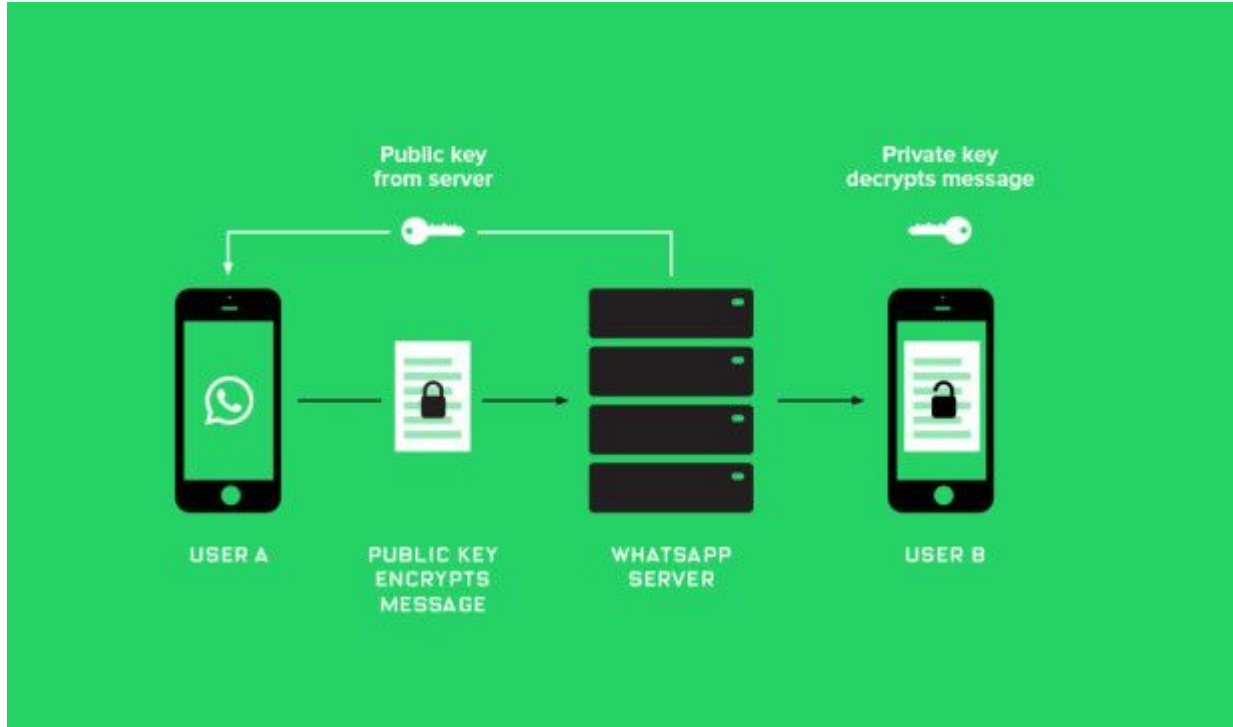
Like most cryptosystems, the security of RSA depends on how it is implemented and used. One important factor is the size of the key. The **larger the number of bits in a key (essentially how long the key is), the more difficult it is to crack through attacks** such as brute-forcing and factoring.

 AUTH_RSA256_d1ff2...1540acc0e97abc2.p12

 RSA256_697bdfd24a...7f5803860a18850.p12



END-to-END Encryption



Медиа, ссылки и докум. Н... >

Избранные сообщения Нет >

Общие группы 1 >

Уведомления >

Не беспокоить Нет >

Медиа По умолчанию >

Шифрование

Сообщения, которые вы отправляете в данный чат, и звонки защищены шифрованием. Нажмите, чтобы подтвердить.



< Данные Подтвердить код б...

Вы, Анна



0 33885 60471 2 022
900 87375 642 33607
5731 69722 75352 32217

Просканируйте код на телефоне другого контакта или попросите его(ее) просканировать ваш код, чтобы обеспечить шифрование ваших звонков и сообщений. Вы также можете сравнить код выше, чтобы подтвердить. [Узнать больше.](#)



Сканировать код

Whatsapp uses the Curve25519 based **algorithm**



One of the most popular messengers in the world was initially released in 2009. In the first version of the app, there were no end-to-end encryption algorithms. For data transmission, WhatsApp uses an open and free protocol called [XMPP](#). It is based on [XML](#) and allows exchanging text messages, audio/video data, and files. In 2012, developers of the app began to work on the end-to-end encryption features. At first, it was only about encrypted text messages. However, starting from 2014 it allows exchanging encrypted text and audio/video messages.



Telegram messenger appeared in 2013. It is available both for mobile and desktop platforms. Users can exchange messages and files of any type. Even though end-to-end encryption is not available by default, you can use this feature by activating the Secret Chat mode. Before sending messages via the transport protocol (HTTP, TCP, UDP), the [MTProto protocol](#) encrypts it.

The Telegram app developers created this end-to-end encryption protocol, and it consists of three parts. The high-level component defines the method that converts API's responses and answers into the binary code. Cryptographic layer defines the method of encryption that will be used before sending the message. The last one, the delivery component, defines the method of message delivery.

During the message preparation, to provide strong end-to-end encryption, Telegram adds the 64-bit key identifier to the body of the message. This identifier defines authorization keys of user and server. Together, they form a 256-bit key and 256-bit initialization vector. This vector is used for message encryption by the AES-256 algorithm. The encrypted message contains the following info: session, message ID, the serial number of the message, and server salt.



This messenger app for iOS was developed in 2011. At the first stage of end-to-end encryption, the message is encrypted by the combination of 2048-bit [RSA](#) public key and 128-bit [AES](#) algorithm. Then on the basis of the [ECDSA](#) (Elliptic Curve Digital Signature Algorithm) algorithm, this messenger creates a signature. After that, users exchange the keys for the end-to-end encryption needs.



Starting from October 4, 2016, Facebook Messenger app allows using end-to-end encryption. This optional feature is available in Secret Conversations mode and includes a timer that shows how long the encrypted messages will remain invisible. Since this option is not available by default, the user cares the data transmission will be safe.

Messaging Apps Security

While messages are E2E-encrypted, backups and metadata are not. WhatsApp **collects** your metadata – **IP address, your name and mobile phone number, location history, your contacts and their phone numbers**, even your **device type**. WhatsApp's new Privacy Policy means that your **metadata will be shared with Facebook**, and there is no way to prevent this. Well, unless you delete the app. However, it is worth mentioning that WhatsApp explained that the **changes are related to optional business features on WhatsApp** and that it **doesn't influence messages with your family or friends in any way**.

Resource: <https://cybercrew.uk/>

WhatsApp's history includes **several hacking attacks**, amongst which one of the victims was none other than **Jeff Bezos**, one of the richest men in the world, **whose phone was hacked through WhatsApp**. In December 2019, WhatsApp confirmed they repaired a security glitch that would enable hackers to access user's data via a **malicious GIF**. Users only had to open their gallery within WhatsApp for the hack to be activated and their content exposed. Luckily, no one was affected by this hazard.

Telegram Security

Telegram uses **client-server encryption** for its personal and group chats (cloud chats), which means that Telegram has access to your messages since it handles the encryption and decryption of them at their servers. For **Secret Chat** and **voice and video calls**, they use the **E2E** Telegram encryption method.

Telegram uses its **MTPROTO**, a custom encrypting protocol developed by **Nikolai Durov** and other Telegram developers. It is based on **256-bit symmetric AES encryption, 2048-bit RSA encryption, and Diffie–Hellman** secure key exchange.

When it comes to **user data**, **Telegram collects** and keeps your **phone number** since it is mandatory for verification of your account, the **username** you choose, and **contact information**. On top of that, they might collect and store for up to 12 months your metadata such as your **IP address, devices and Telegram app you've used, history of Telegram username changes, etc.** They state that they might reveal your IP address and phone number if asked by relevant authorities, so if you want to hide your IP address, consider using a VPN.

Same as WhatsApp, Telegram has been **hacked several times**. One of the times was in 2019, during **Hong Kong protests** when activists reported security breaches that enabled law enforcement to track them. These security violations inspired Telegram to add a new option for users, allowing them to **hide their phone number from everyone**. In the mid-2020, Telegram was disclosed to a security breach in which users' phone numbers and Telegram IDs were exposed on the darknet. The company stated that 60% of leaked data was incorrect, and 84% was outdated, 70% of the leaked accounts were based in Iran, and 30% were from Russia.